# Module 3 - time-to-progression model (module3/model.py)

X
real baseline
image only
(380 × 380)

**EfficientNet-B4**
**(ImageNet init,**
**fine-tuned)**

1792-d

concat
1808

Linear 1808 → 128
ReLU
Dropout 0.3
Linear 128 → 2
softplus

shape k
scale λ

S(t) = exp(−(t/λ)^k)

P(progression
within 2 years)
= 1 − S(2)
+ survival curve

LBS stratum
low / med / high
(from Module 1)

Embedding
3 → 8

baseline grade
ICDR 0–4
(from Module 1)

Embedding
5 → 8

Training (Tianjin, baseline → fixed 2-year follow-up):
event e = 1 if follow-up grade > baseline grade, observed only at t = 2 years
current-status Weibull NLL = −[ e·log(1 − S(2)) + (1 − e)·log S(2) ]
AdamW lr 1e-4 (wd 1e-2), first 6 blocks frozen, augmented, 30 epochs
patient-level train / val / test 70/15/15; best epoch = lowest val NLL; reported on test